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Sustainability of rice farming based on eco-farming to face food security and climate change: Case study in Jambi Province, Indonesia

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Abstract

The objectif of this study was to analyze index and sustainability status of rice farming carried out by farmers in Jambi Province, Indonesia. The analysis used modification of Rapfish method by using multidimensional scaling (MDS) technic [1]. The analysis of sustainability of rice farming in multidimensional revealed that rice farming carried out by farmers is less sustainable, with the sustainability index is 41.96. Indexs to the dimensions of ecological, economic, socio-cultural and technological sustainability are 48.25: 40.83: 40.97, and 16.46 respectively. This indexs indicate that ecological, economic, and socio-cultural dimensional were less sustainable, and index to the dimensions of technological was not sustainable. According to attribute leveraging analysis shows that there are some aributes as a leverage factor, ie: rice productivity, the usage of organic fertilizers, organic pesticides, and land capability class (for the ecological dimension); the percentage of income to total revenue, the percentage of farmers who live below the poverty line, farmer exchange, government subsidies, availability production facilities, comparative advantage, and the development of rice prices (economic dimension); farmers motivation, the average age of farmers, the adoption rate of technology, the time required for rice farming, rice consumption growth and government policy (socio-cultural dimensions); post-harvest technology, the availability of improved seed, the amount of rice milling, and how to eradicate the pest management (technological dimension).

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Keywords: Rice farming, sustainability index, stainability status, rapfish, eco-farming

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1. Introduction

One of the important themes in the debate around how to ensure global food security concerns the capacity of the planet to support its growing population. The human population, particularly in developing Asian and African countries, is growing at an exponential rate [2]. This problem driver effort to meet their basic needs. The conventional farming try to meet the need of growing population by using natural resource. Human interaction with the environment to meet their needs has led to the depletion of natural resources leading to massive environmental degradation, especially in developing countries. Reed, say that the problem of environmental degradation has become a common phenomenon in rural communities in developing countries, especially in societies that are still faced with the problem of meeting basic needs and poverty [3].

Indonesia that known as an agricultural country has not been able to realize the strong and healthy agricultural system, both as food and employment providers. It can be seen in the fact that the national food security is still lacking with the indicators are still high imports volume of strategic important commodities such as rice. Indonesia is still the highest rice importer to fulfil the national rice sufficiency. This is because of the low productivity of paddy and higher average per capita rice consumption [4]. Rice production is also give impact on the environment due to pollution and environmental degradetion from the use of chemical fertilizers and pesticides, as well as greenhouse gas emissions. On the other hand, more than 20% of Indonesia's population has engaged in rice farming. Hence it is necessary to analyze the sustainability of rice farming and create scenarios to achieve food security as well as wise to the environment.

2. Method

This study was conducted in the three counties of Jambi Province to analize the characteristics of farmers. Judging the sustainability of rice farming used multidimensional scaling (MDS) is called the Rap-Paddy. This method is developed by doing modification to the Rap-Fish (The Rapid Appraisal of the Status of Fisheries) method, which has been used to assess the sustainability status of fisheries [1]. Rap-Paddy analysis will result the status and sustainability index of rice farming. The index value of the sustainability of rice farming is called as SI-Paddy (Sustainability Index of Paddy). Monte Carlo analysis helps to see the effect of errors in making the score on each attribute.

3. Result and Disscussion

Jambi province is located in Central Sumatera, Indonesia, geographically located between 0º45' south latitude - 2º45' south latitude and 101º10' east longitude - 104º44' east longitude. Jambi province has a varied topography ranging from altitude 0 m above sea level in the eastern part of the level, up to above 1,000 m above sea level in the west. The climate belongs to the A type as a classification based on the Schmidt-Ferguson, with an average rainfall ranges from 2600 mm/year. The rainy season occurs in almost every month throughout the year [5].

The role of rice in the construction can be seen from its performance in the past five years (2005-2009) have increased 0.28%. The rice production increased 2.79%[6]. This increase is very small and very out of balance with the increase of population (2% per year) and rice consumption.

Conditions of the most farmers are reflects to the life characteristic was not supported by educational level. Dominated by education level of farmers had not completed and graduated elementary school about 46.2%. This suggests that the level of education pursued by the farmers is still relatively low, so that it can lead to the adoption of technologies by farmers was not optimal.

Farming system carried out by farmers is still conventional, which is characterized by the use of chemicals is quite high. About 23.7% of farmers are still letting the rest of the rice straw rotted in the fields. This is one of the causes of methane emissions from rice farming. Methane is one of gas that forming greenhouse gas, has led to global warming. Characteristics of rice farming shows that rice farming requires the time allocation per day is high enough, where 63.9% allocate time for rice farming more than 9 hours per day, and 36.1% of the farmers allocate time 6-8 hours per day. This condition indicates that if the farmer has come down to the fields, the farmers do not have the opportunity for other businesses. This led to the limitations of farmers in developing business diversification.

Rap-Paddy analiysis reveald that the SI-Paddy is 41.96. This value reflects that rice farming carried out by farmers at the household level less sustainable in multidimensional. This value is obtained based on an assessment of the 42 attributes that are included into the four dimensions, ie the ecological dimension (11 attributes), economics dimension (11 attributes), socio-cultural dimension (11 attributes), and technology dimension (9 attributes). Sustainability index value for each dimension, the value of stress, and R^2 can be seen on Table 1.

Table 1. Result of MDS, Monte Carlo and Statistics analysis

Dimensions	Sustainability Index		Difference	Statistics		Iteration
	MDS	Monte Carlo		Stress	R^2	
Multidimension	41,96	41.97	0,01	0,13	0,96	2
Ecological Dimension	48,25	48,26	0,01	0,15	0,95	2
Economics Dimension	40,83	40,91	0,08	O,14	0,95	2
Socio-cultural Dimension	40,97	40,99	0,02	0,14	0,95	2
Technological Dimension	16,46	16,45	0,01	0,14	0,95	2

Table 1 show that the value of multidimensional stress test results is quite low at 0.13, and the coefficient of determination (R^2) reached 0.96, it is indicating that the quality analysis is good. The same is obtained from the sustainability analysis of the ecological, economics, socio-cultural and technological dimension, where the stress value is quite low (<0.25) and the coefficient of determination (R^2) reached 0.95. The both of parameter indicate that all attributes are used quite well in explaining the index and sustanability status of rice farming.

The difference sustanability index value between the MDS and the Monte Carlo is quite small, this indicates that the error in making scores of each attribute is relatively small, the variation of scores due to differences in opinion are relatively small, the analysis performed repeatedly stable, and error data entry and data loss can be avoided. Overall, these results suggest that Rap-Paddy analysis using MDS method to determine the sustainability of rice farming had high levels of trust. Multidimensional sustainability index shown on Figure 1.

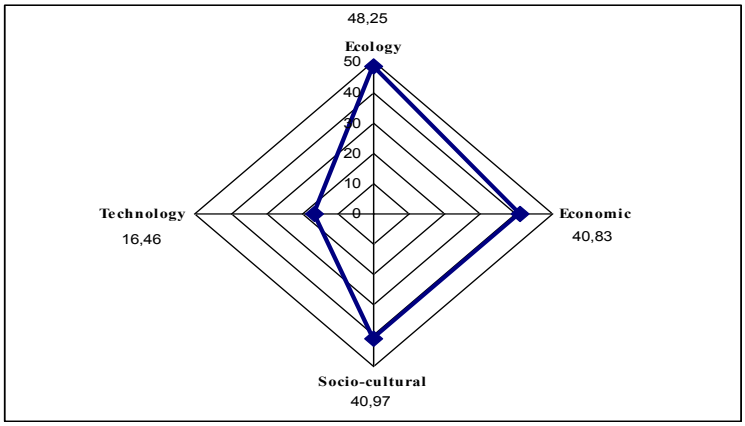

Figure 1. Kite diagram of four rap-paddy evaluation fields showing status score of rice farming

Rap-Paddy analysis showed that the index of sustainability for ecological dimension is 48.25 (Figure 2), this value indicates that rice farming is done by farmers less sustainable. Sustainability index of ecological dimension and sensisitive attributes of the farming system shown on Figure 2.

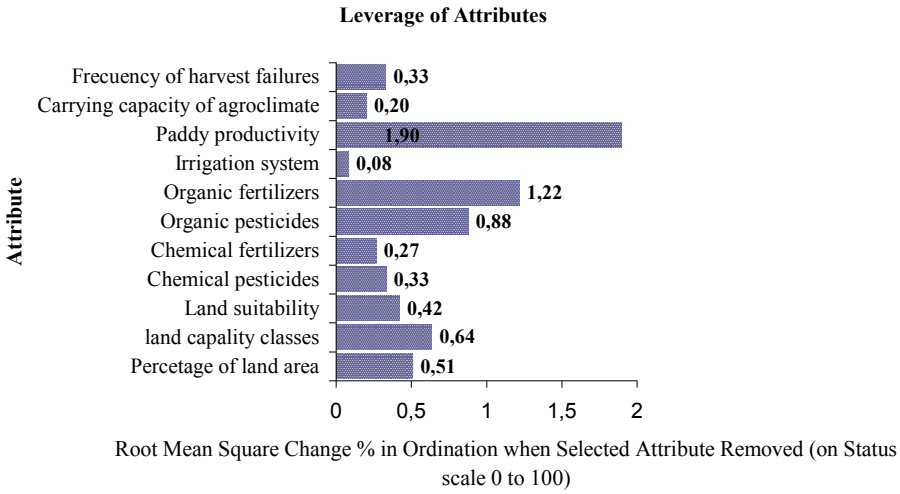

Figure 2. Leverage of eleven ecological attributes on sustainability score

Figure 2 show that the sensitive attributes in determining the sustainability of rice farming is a). paddy productivity b). the use of organic fertilizers, b). the use of organic pesticides, and d) land capability class. Paddy productivity averaged only 3 tons per hectare is due basically rice plants are less suitable for tropical regions only have rainy and dry, with an average of sunlight per day is low, less than 12 hours per day throughout the year. This is contrast to sub-tropical countries, when the summer season get 16-20 hours of sunlight per day. Paddy is kind of C4 plants, if planted in summer, the production can reach 9 tons per hectare, for a long time photosynthesis [7]. Te low productivity of paddy in Jambi also caused by

the leveling off the land as a result of the usage of chemical fertilizers. However, this low paddy productivity can be improved by the provision of organic fertilizer from cow manure.

The usage of organic fertilizers is one of the sensitive attributes bacause of existing conditions using only a small proportion of organic fertilizer. This is because famers are fears of a decline in rice production. Farmers think that by using organic fertilizers produced rice yield is lower than the use of chemical fertilizers. Other causes less use of organic fertilizers for farmers' knowledge and capacity in processing animal waste into organic fertilizer is still low.

The application of pesticide/herbicide organic in rice farming has not been performed, because they do not have the knowledge and skills in making pesticide organic, plant-based pesticides are also not yet commercially available in the market. The usage of pesticides/ herbicides chemicals by farmers that has been pollution and environmental degradation.

To improve the status of sustainability of rice farming in the ecological dimension, the future need to be made to improve rice productivity through the use of organic fertilizers and pesticides by increasing the knowledge, capacity and skills of farmers to process and use of organic fertilizers and pesticides.

Sensitive attributes that affect the sustainability of rice farming on the economic dimension is a). percentage of income from rice farming to total income, b). percentage of rice farmers who live below the poverty line, farmer exchange c). government subsidies, d). availability of the means of rice production, d). comparative advantage and development of rice prices (Figure 3).

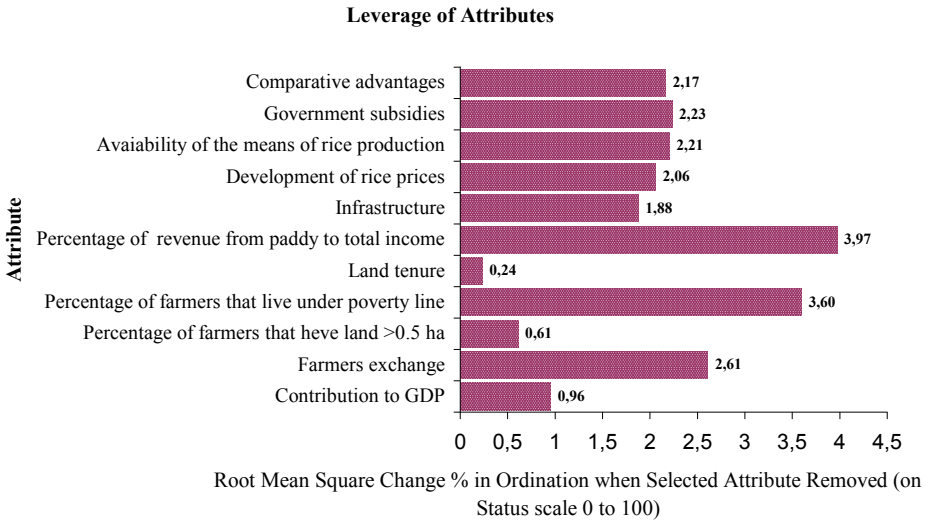

Figure 3. Leverage of eleven economic attributes on sustainability score

Percentage of income from rice farming to total income of farmers is still low. This is because production in rice farming inputs (chemical fertilizers and chemical herbicides) is quite high, and low productivity on the other side so the rice does not provide significant advantages over total revenue. But most of the rice farmers while maintaining the grounds as a stock farming household food needs. In addition, to ensure the sustainability of rice farming in the economic dimension should be supported by the price of rice that can provide sufficient income to farmers.

Paddy Rap-analysis on the social dimension of culture showed that the sustainability index of 40.97. This value shows that the socio-cultural, rice production is also less sustainable. Attributes are sensitive to

the socio-cultural dimensions as shown in Figure 4. The sensitive attributes in determining the sustainability of rice farming is on the socio-culture dimension is a). motivation in farming, b). the average age of farmers, c). level of technology adoption, and d). outpouring of time per day in rice farming. One dimension of the critical success rice farming is a social and cultural conditions of farmers. Social and cultural conditions are very relevant to the circumstances of farmers in the daily conduct of farming. Socio-cultural farm community will experience a shift, especially in terms of technology adoption is determined by age level, and motivation of farmers to be able to thrive in the management of rice farming.

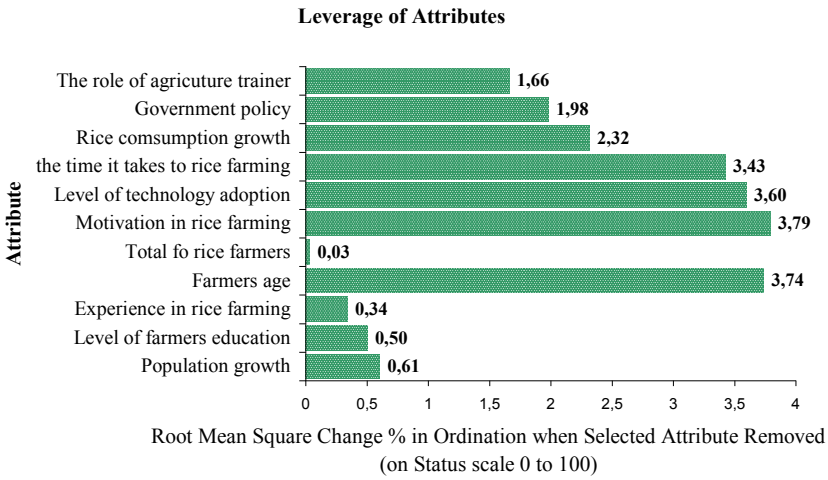

Figure 4. Leverage of eleven socio-cultural attributes on sustainability score

Paddy Rap-analysis on the technology dimension showed that the sustainability index of 16.46. This value shows that rice production in technology dimension is not sustainable. Attributes are sensitive to the technology dimensions as shown on Figure 5.

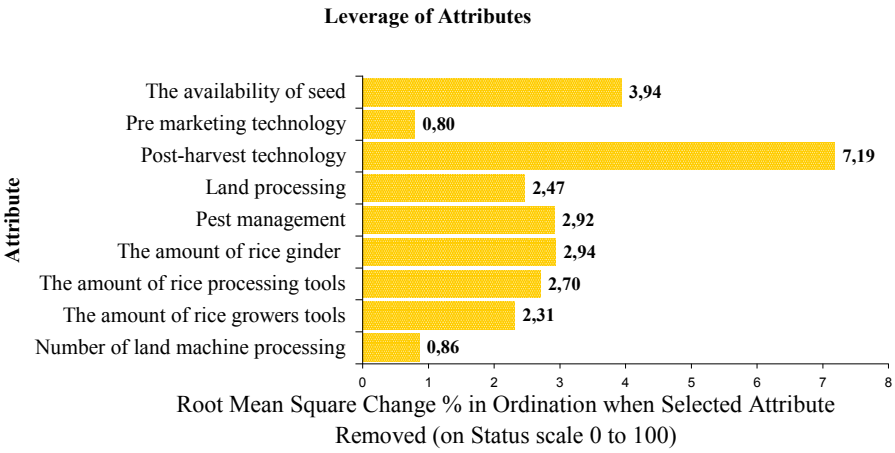

Figure 5. Leverage of nine technological attributes on sustainability score

Figure 5 shown that the sensitive attribute in determining the sustainability of rice farming is a post-harvest technology and the availability of improved seed. This indicates that the seed or rice varieties suitable for tropical climate affects the sustainability of rice farming. According Thayib, there is often a false perception in environmental rules, including the selection of rice seeds. Rice is grown in sub-tropical regions, species of *Oryza sativa* japonica or *forma Sinica*, a long day so that plants are not suited to the tropics [7]. Post-harvest technology is also very influential as they relate to the quality of the product before marketing.

Closing

Characteristics of farmers is its low level of education, the age is still productive, have a fairly high land. The rice farming system is still conventional, characterized the high use of chemical fertilizers and pesticides. Status of the sustainability of rice farming in multidimensional less sustainable with sustainability index 41.96. The sustainability index ecological, economic, socio-cultural and technologcal dimensions are 48.25, 40.83, 40.97 and 10.46 respectively. The value indicate that the farming system is less sustainable. For the future, Indonesia have to develop the food forest to food security and to solve the environment degradation.

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